Spatially-Correlated Distributionally Robust Scheduling for Carbon-Aware Data Centers

Status Report — Revised Feasible Set and a Second-Grid Generalization

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6 June 2026

Abstract

We report this period's progress on the distributionally robust optimization (DRO) approach to carbon-aware data-center scheduling. The central question is whether modeling the *spatial correlation* of carbon intensity across electrically coupled regions improves a robust schedule. We (i) replaced a previously dominant aggregate power cap with a more defensible feasible set — adding a job deadline constraint and a temperature-coupled cooling (PUE) constraint driven by a new external temperature dataset — and (ii) transported the *identical* methodology to a second, independent grid (Iberia and France) to test generalization. Using a falsification test (the schedule fitted to the empirical joint covariance versus one with all cross-region structure destroyed), evaluated by out-of-sample Conditional Value-at-Risk on a held-out year under strict pre-registration, we find that spatial correlation yields **no robust scheduling value** in either grid: the worst-case tail emissions change by at most 0.06% across all constraint regimes and flexibility levels. We demonstrate the null is *substantive* (the optimizer is not frozen) rather than a mechanical artifact, that it survives a shrinkage and residual-space robustness battery, and — notably — that it holds even though Spain and Portugal have an empirical carbon-intensity correlation of ≈ 0.85, higher than any California pair. The methodological contribution is a clean, replicated negative result with a mechanistic explanation.

1 Introduction and scope of this period

The project extends a Wasserstein-DRO framework for carbon-aware scheduling by treating carbon intensity as a stochastic vector across multiple co-regional clusters, so that *cross-region* (spatial) correlation can in principle inform the schedule. This report covers two work items:

1. **A revised feasible set.** The earlier model relied on a single aggregate power cap that dominated the geometry of the feasible region. We removed it and rebuilt the operational constraints from more physically grounded components — a deferral-deadline constraint and a temperature-coupled Power-Usage-Effectiveness (PUE) constraint — and re-validated the central experiment across three constraint regimes of increasing tightness.
2. **A second, independent grid.** To assess whether the conclusion is specific to California, we applied the unchanged methodology to Spain, Portugal and France. Spain and Portugal share the MIBEL day-ahead market and are tightly coupled, directly addressing the requirement that the regions be electrically coupled to be of interest.

The optimization method itself — the Mahalanobis–Wasserstein second-order cone program (SOCP) of Algorithm 2b — is held fixed throughout; only the feasible set and the region set vary.

2 Methodology

2.1 The scheduling problem

We schedule compute across R regions over a $T = 24$-hour day. The decision variable is

$$x \in \mathbb{R}^{R\times T}_{\geq 0}, \qquad x_{r,t} = \text{compute power in region } r \text{ at hour } t \text{ [MW]}.$$

The carbon intensity of each region–hour, $\rho_{r,t}$ (in gCO_2eq/kWh), is uncertain. The carbon cost of a schedule is linear in ρ:

$$\langle \rho, x\rangle = \sum_{r=1}^{R}\sum_{t=0}^{T-1} \rho_{r,t}\, x_{r,t}.$$

2.2 Distributionally robust objective

Rather than trust a single forecast, we minimize the worst-case expected cost over a Wasserstein ambiguity ball $\mathcal{B}_\varepsilon(\hat{\mathbb{P}})$ of radius $\varepsilon \geq 0$ centered on the empirical distribution $\hat{\mathbb{P}}$ of the daily carbon field:

$$\min_{x\in\mathcal{X}} \sup_{\mathbb{Q}\in\mathcal{B}_\varepsilon(\hat{\mathbb{P}})} \mathbb{E}_{\mathbb{Q}}\big[\langle \rho, x\rangle\big]. \tag{1}$$

For a cost linear in ρ and a type-2 Wasserstein ball whose ground metric is the Mahalanobis distance induced by the empirical covariance $\hat{\Sigma}$, the inner supremum admits the closed form of Wasserstein-DRO duality [1]: the worst case equals the empirical-mean cost plus ε times the Mahalanobis norm of the schedule. Problem (1) reduces to

$$\boxed{\min_{x\in\mathcal{X}} \quad \underbrace{\langle \bar{\rho}, x\rangle}_{\text{mean carbon}} \quad + \quad \underbrace{\varepsilon\sqrt{x^\top \hat{\Sigma}\, x}}_{\text{robustness penalty}}} \tag{2}$$

where $\bar{\rho} \in \mathbb{R}^{R\times T}$ is the empirical mean field and $\hat{\Sigma} \in \mathbb{R}^{RT\times RT}$ is the empirical covariance of the flattened daily field $\operatorname{vec}(\rho)$, using the fixed row-major convention $\operatorname{vec}(x)_{rT+t} = x_{r,t}$. The penalty $\sqrt{x^\top \hat{\Sigma} x}$ is the standard deviation of the schedule's carbon cost under $\hat{\Sigma}$; the radius ε controls the strength of the hedge, with $\varepsilon = 0$ recovering the deterministic problem on $\bar{\rho}$.

2.3 Where spatial correlation enters

Partition $\hat{\Sigma}$ into $T \times T$ blocks indexed by region pairs,

$$\hat{\Sigma} = \begin{pmatrix} \Sigma_{11} & \cdots & \Sigma_{1R} \\ \vdots & \ddots & \vdots \\ \Sigma_{R1} & \cdots & \Sigma_{RR} \end{pmatrix}, \qquad \Sigma_{rs} = \operatorname{Cov}\big(\rho_{r,\cdot},\, \rho_{s,\cdot}\big) \in \mathbb{R}^{T\times T}. \tag{3}$$

The diagonal blocks Σ_{rr} are within-region temporal covariances; the off-diagonal blocks Σ_{rs} ($r \neq s$) are exactly the cross-region (spatial) covariance. Through the quadratic form

$$x^\top \hat{\Sigma} x \;=\; \sum_{r} x_r^\top \Sigma_{rr}\, x_r \;+\; \sum_{r\neq s} x_r^\top \Sigma_{rs}\, x_s, \tag{4}$$

the off-diagonal blocks are the *only* channel through which spatial correlation can change the schedule. The experiment in §2.6 manipulates precisely these blocks.

2.4 Second-order cone reformulation

We factor the regularized covariance by its Cholesky decomposition $\hat{\Sigma} = LL^\top$ with L lower triangular. Then $\sqrt{x^\top \hat{\Sigma} x} = \sqrt{x^\top LL^\top x} = \|L^\top \operatorname{vec}(x)\|_2$, so (2) becomes

$$\min_{x \in \mathcal{X}} \langle \bar{\rho}, x \rangle + \varepsilon \left\| L^\top \operatorname{vec}(x) \right\|_2, \tag{5}$$

which, via its epigraph $\min \langle \bar{\rho}, x \rangle + \varepsilon s$ subject to $\|L^\top \operatorname{vec}(x)\|_2 \le s$, is a convex second-order cone program solved to global optimality (CLARABEL). The orientation $L^\top$ is essential: it yields $x^\top (LL^\top) x = x^\top \hat{\Sigma} x$, whereas $\|L \operatorname{vec}(x)\|_2$ would compute the different form $x^\top L^\top L\, x$; the implementation pins the correct orientation with a post-solve numerical cross-check.

2.5 The revised feasible set $\mathcal{X}$

Every constraint below is linear, so (5) remains an SOCP.

C0 — per-cell ceiling (kept). $0 \le x_{r,t} \le \bar{x}_{r,t}$; a site cannot exceed its installed capacity ($\bar{x} = 50$ MW).

C2 — flexible/inflexible split (kept). A fraction $\alpha_r \in [0,1]$ of each region's daily work W_r is inflexible on a fixed intraday shape $p_{r,t}$ ($\sum_t p_{r,t} = 1$); the remainder is flexible:

$$x_{r,t} = \alpha_r W_r\, p_{r,t} + x^{\text{flex}}_{r,t}, \quad x^{\text{flex}}_{r,t} \ge 0, \quad \sum_t x^{\text{flex}}_{r,t} = (1-\alpha_r)\, W_r. \tag{6}$$

Here α is the master flexibility dial: lower α leaves more movable work and thus more room for a spatial effect. We sweep $\alpha \in \{0.30, 0.50, 0.75\}$ and fix utilization so $W_r = 0.80\, \bar{x}\, T = 960$ MWh.

C3 — inter-hour ramp (kept). $|x_{r,t} - x_{r,t-1}| \le \Delta_r$ for $t = 1, \dots, T-1$, with $\Delta_r = 15$ MW/h. This makes the naive "cheapest-hours-first" schedule infeasible.

C1 — aggregate power cap (dropped). The previous constraint $\sum_r x_{r,t} \le P_{\max}$ is removed; it was a single global knob that flattened the feasible polytope.

3a — windowed-demand / deferral deadline (new). For a window $[\tau_1, \tau_2]$ and fraction γ, a γ-share of each region's flexible work must clear inside the window:

$$\sum_{t=\tau_1}^{\tau_2} x^{\text{flex}}_{r,t} \ \ge\ \gamma\,(1-\alpha_r)\, W_r. \tag{7}$$

This is an aggregate deadline (a floor on flexible work completed by the window's end), not a per-job service-level agreement. We use the morning window $[0,7]$ with $\gamma = 0.20$.

3b — temperature-coupled thermal/PUE constraint (new). Usable IT power is limited by effective (wall-plug) power, the IT power inflated by the facility PUE, which rises with outdoor temperature through a piecewise-linear "hockey stick":

$$\text{PUE}(T) = \text{PUE}_0 + \kappa\, \max\big(T - T_{\text{set}},\, 0\big), \qquad \text{PUE}\big(T^{\text{air}}_{r,t}\big)\, x_{r,t} \ \le\ \bar{P}. \tag{8}$$

Because the air temperature $T^{\text{air}}_{r,t}$ is data (a fixed field, never re-estimated per fold nor read from the test set), PUE$(\cdot)$ is a known constant per cell and (8) is linear — it tightens the ceiling to $\bar{P}/\text{PUE}(T^{\text{air}}_{r,t})$ in hot hours. We use $\text{PUE}_0 = 1.10$, $\kappa = 0.015\,/°\text{C}$, $\bar{P} = 57$ MW, and economizer set-point $T_{\text{set}} = 20°\text{C}$ (California) or 14°C (the milder Iberian field; see §5).

2.6 The falsification test: joint versus shuffled covariance

To isolate the value of *spatial* correlation specifically, we feed the scheduler two covariances that differ in exactly one respect. The joint covariance $\hat{\Sigma}^{\text{joint}}$ is the full empirical covariance (3). The shuffled covariance $\hat{\Sigma}^{\text{shuf}}$ zeros all cross-region blocks while preserving the within-region blocks:

$$\left(\hat{\Sigma}^{\text{shuf}}\right)_{rs} = \begin{cases} \Sigma_{rr}, & r = s, \\ \mathbf{0}_{T\times T}, & r \neq s. \end{cases} \tag{9}$$

This preserves each region's marginal distribution and within-region temporal autocorrelation, and destroys only cross-region dependence. Hence comparing the scheduler under $\hat{\Sigma}^{\text{joint}}$ versus $\hat{\Sigma}^{\text{shuf}}$ isolates exactly the value of spatial correlation. If that structure is useful, the joint-fitted schedule should achieve lower out-of-sample tail cost. The test is a *falsification*: it is built to detect a spatial benefit if one exists, so a null is informative.

2.7 Evaluation protocol

Risk metric. A schedule x is scored on each held-out test day i by its realized emissions $e_i = \langle \rho_i, x\rangle$. The objective is the Conditional Value-at-Risk at level $\beta = 0.95$ — the mean of the worst 5% of days:

$$\text{CVaR}_\beta\big(\{e_i\}_{i=1}^N\big) = \frac{1}{\lceil(1-\beta)N\rceil}\sum_{i\in\mathcal{T}_\beta} e_i, \qquad \mathcal{T}_\beta = \{\text{indices of the largest } \lceil(1-\beta)N\rceil\ e_i\}. \tag{10}$$

We report the *spatial gap* $\text{gap} = \text{CVaR}_{0.95}(\text{shuf}) - \text{CVaR}_{0.95}(\text{joint})$, so a positive gap means the joint covariance helped; we express it as a percentage of the shuffled CVaR so the two grids are comparable.

Selecting ε. For each arm separately, ε is chosen by 5-fold *blocked* (contiguous, non-shuffled) time-series cross-validation on 2021–2024 over the grid $\{0, 0.1, 1, 10, 100, 1000\}$, minimizing mean validation CVaR; the selected $\varepsilon^\star$ is refit on the full training set and evaluated once on the untouched 2025 test set.

Uncertainty. A 95% confidence interval on the gap is obtained by a paired bootstrap: test-day indices are resampled with replacement (the same indices for both arms), the gap recomputed on each of 1000 resamples, and the 2.5/97.5 percentiles reported; a gap is *detectable* only if the interval excludes zero.

Pre-registration. For every experiment we (i) ran a dry-run gate performing the full cross-validation without touching 2025; (ii) commit-locked the configuration in version control before any test-set evaluation; (iii) read the test set exactly once. This removes the analyst degrees of freedom that could bias the outcome.

3 Experimental design

The joint-versus-shuffled comparison is run across three nested constraint regimes so that the value of spatial correlation is reported as a function of operational tightness (Table 1). Two region sets are studied with *identical* method, metric, cross-validation, bootstrap and table format; only what must differ does.

Table 1: The three nested constraint regimes (each adds to the previous).

Regime	Active constraints	Role
R3 (reference)	C0 + C2 + C3	post-cap baseline (anchor)
R1 (lean)	R3 + deadline (3a)	headline regime; no weather assumptions
R2 (full)	R1 + thermal (3b)	tightest, most assumption-laden

Data. Carbon intensity is hourly life-cycle intensity from Electricity Maps (academic tier), 2021–2025. Temperature is hourly 2 m air temperature from the Open-Meteo ERA5 archive, one load-weighted point per zone (California: Fresno, Sacramento, Los Angeles, Las Vegas; Iberia: Madrid, Lisbon, Paris). Both panels are built on a common local clock, train on 2021–2024 and test on 2025.

4 Results

4.1 Case A — California / Nevada ($R = 4$)

Zones CISO, BANC, LDWP and NEVP. Test-set $\mathrm{CVaR}_{0.95} \approx 1.59 \times 10^6$ $\mathrm{gCO_2}$. Table 2 reports the spatial gap. The largest benefit anywhere is +0.058%; the cells flagged detectable at $\alpha = 0.75$ are economically meaningless (0.0017%, near-identical schedules).

Table 2: California: spatial gap $\mathrm{CVaR}_{0.95}(\mathrm{shuf}) - \mathrm{CVaR}_{0.95}(\mathrm{joint})$ by regime and α. Positive favors the joint covariance.

Regime	α	gap (%)	95% CI ($\mathrm{gCO_2}$)	verdict
R3	0.30	+0.0583	[+211, +1942]	detectable
R3	0.50	+0.0357	[−532, +2177]	null
R3	0.75	−0.0122	[−581, +488]	null
R1	0.30	+0.0164	[−118, +831]	null
R1	0.50	+0.0271	[−359, +1577]	null
R1	0.75	+0.0017	[+13, +41]	detectable*
R2	0.30	+0.0185	[+34, +693]	detectable
R2	0.50	+0.0262	[−267, +1366]	null
R2	0.75	+0.0017	[+13, +41]	detectable*

* Statistically detectable but 0.0017% of CVaR — practically null.

4.2 Case B — Iberia + France ($R = 3$)

Zones ES, PT, FR. Test-set $\mathrm{CVaR}_{0.95} \approx 0.40 \times 10^6$ $\mathrm{gCO_2}$ (lower than California because French intensity is low: mean ≈ 50 versus ES/PT ≈ 160 $\mathrm{gCO_2eq/kWh}$). Table 3 reports the gap; eight of nine cells are practically null.

Table 3: Iberia + France: spatial gap $\mathrm{CVaR}_{0.95}$ by regime and α.

Regime	α	gap (%)	95% CI (gCO_2)	verdict
R3	0.30	+0.0000	$\varepsilon^\star$=0, schedules identical	null
R3	0.50	−0.0000	$\varepsilon^\star$=0.1, ≈ identical	null
R3	0.75	+0.0210	[−108, +226]	null
R1	0.30	−0.0017	[−843, +678]	null
R1	0.50	+0.0151	[−18, +129]	null
R1	0.75	+0.3211	[+746, +1870]	detectable†
R2	0.30	−0.0075	[−563, +495]	null
R2	0.50	+0.0000	[−0.004, +0.013]	null
R2	0.75	−0.0155	[−152, +19]	null

† A cross-validation selection artifact; see §4.3.

4.3 The apparent exception and why it does not survive

The Iberian R1/α=0.75 cell shows a raw +0.32% gap with a confidence interval excluding zero — the single largest raw gap in either study. It arises *only* because cross-validation hands the two arms different radii ($\varepsilon^\star_{\text{joint}} = 1$ but $\varepsilon^\star_{\text{shuf}} = 10$), i.e. it compares two schedules at different penalties rather than measuring an informational advantage. When the comparison is made clean — residual-space estimation, under which both arms select $\varepsilon^\star = 1$ — the sign flips (Table 4).

Table 4: The R1/α=0.75 cell across estimators. The raw effect reverses under clean (equal-$\varepsilon^\star$) comparison.

Estimator	$\varepsilon^\star$ joint/shuf	gap (%)	clean comparison?
Sample covariance	1/10	+0.321	no — $\varepsilon^\star$ diverge
Ledoit–Wolf shrinkage	1/10	+0.321	no — $\varepsilon^\star$ diverge
Seasonal residual	1/1	−0.032	yes
AR(1) residual	1/1	−0.037	yes

Under both clean estimators the joint covariance is marginally *worse*, and they agree in sign; our pre-registered "seasonal and AR(1) must agree" rule therefore records no beneficial spatial effect. The apparent exception is a selection artifact, caught by the robustness machinery designed to catch it.

Verdict. In both grids the gap is a few hundredths of a percent of CVaR essentially everywhere; neither shows an economically meaningful, robust spatial benefit. **Iberia replicates the California null; the more tightly coupled MIBEL grid did not surface spatial value.**

5 Reliability of the result

Substantive, not mechanical (the Goldilocks check). A null is meaningful only if the optimizer *could* have responded to spatial structure. We verify the feasible set is not frozen by measuring how far the deterministic optimum reallocates as ε grows, $\mathrm{move}(\varepsilon) = \|x(\varepsilon) - x(0)\|_1$ as a fraction of total work. Across all nine regime×α cells the optimizer retains substantial freedom — 24–32% in California and 9.5–18% in Iberia — so the null reflects a genuine absence of exploitable spatial structure. By design every constraint is active but not dominant (in R2 the thermal cap binds $\approx$ 22–26% of cells; the deadline binds for most regions), present and realistic without removing the optimizer's freedom.

Robustness battery. (i) *Ledoit–Wolf shrinkage* [3]: re-running with the shrinkage covariance reproduces every gap to ≈ 4 significant figures, so the result is not an artifact of a noisy sample covariance. (ii) *Residual-space*: re-estimating the covariance from the forecast residuals of a seasonal and an AR(1) baseline (keeping the mean field and scored emissions on raw intensity), with a pre-registered agreement rule, yields no beneficial spatial effect in either grid.

Replication across independent grids. Two grids differing in geography, climate, generation mix and coupling structure (CAISO/Nevada and MIBEL/France) give the same conclusion. A replicated null under a falsification test is substantially stronger evidence than any single region.

Reproducibility. Every value here is read from committed result files; the pipeline carries 149 unit tests, and the configuration was version-locked before the test set was scored. The single re-calibrated parameter across the two grids was the thermal set-point ($20 \to 14$°C), because the milder Iberian annual-mean field would otherwise leave constraint (8) inactive; the headline regime R1 carries no thermal assumption, so the conclusion does not depend on this choice.

6 Interpretation: why this is the expected outcome

High *marginal* correlation does not imply exploitable *robust-scheduling* value. In Iberia the Spain–Portugal carbon-intensity correlation is ≈ 0.85, higher than any California pair (0.43–0.77), yet the joint covariance buys nothing over the block-diagonal one. Two mechanisms explain this. First, cross-validation selects small radii ($\varepsilon^\star \in \{0, \dots, 1\}$), so the robustness term in (5) is a minor correction on top of the mean-cost term shared identically by both arms; the schedule is driven mostly by where carbon is cheap on average. Second, in the quadratic form (4) the within-region (diagonal) terms dominate for a single fixed schedule concentrated in each region's cheap hours, so zeroing the off-diagonal blocks Σ_{rs} changes the penalty — and hence the optimal schedule and its tail cost — by very little. Spatial structure would have the most leverage if the operator could continuously reallocate *between* regions in response to a realized joint shock; the present design commits to one representative schedule, exactly the setting in which spatial correlation has the least purchase.

7 Limitations and next steps

- **One representative schedule.** A rolling day-ahead formulation that re-optimizes each day against updated forecasts is the most likely route to a genuine spatial benefit and the highest-value next experiment.
- **Common-clock convention.** Portugal (WET) is scheduled on the Spain/France (CET) clock, a one-hour offset; this cannot bias the joint-versus-shuffled comparison (it perturbs both arms identically) and is documented.

- **Reporting clarity.** The detectability flag will be supplemented with a practical- significance threshold so trivially small but CI-tight cells are not misread.
- **Publication track.** Given the replicated null, the planned vine-copula extension may pivot toward characterizing *when* spatial structure matters (metric, ambiguity set, or the adaptive setting above) rather than presuming it does.

8 Conclusion

On a defensible feasible set, under a falsification test evaluated by out-of-sample CVaR with strict pre-registration, spatial correlation of carbon intensity provides no robust scheduling value — a result that replicates across two independent grids and survives a shrinkage and residual-space robustness battery, and which we show to be substantive rather than a mechanical artifact. The contribution this period is a clean, replicated negative result with a mechanistic explanation, and a clear next experiment (adaptive re-optimization) under which a positive effect, if any exists, would be most likely to appear.

References

[1] P. Mohajerin Esfahani and D. Kuhn. Data-driven distributionally robust optimization using the Wasserstein metric. *Mathematical Programming*, 171:115–166, 2018.

[2] G. Hall et al. Distributionally robust carbon-aware scheduling with virtual capacity curves. arXiv:2410.21510, 2024.

[3] O. Ledoit and M. Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.

[4] A. Radovanović et al. Carbon-aware computing for data centers. *IEEE Transactions on Power Systems*, 38(2):1270–1280, 2022.

A Reproducibility

Repository `ma731/Capstone-in-Operations-Research`, branch `main`. Result files: `results/taskA_regimes_2026-` (California) and `results/es_pt_fr_regimes_2026-06-05*.csv` (Iberia + France, including the `_lw`, `_seasonal` and `_ar1` robustness variants). Detailed working notes: `docs/progress_note_v13.md` and `docs/progress_note_v14.md`. Mathematics: `thesis/full_formulation.md` and `thesis/constraints_expla` The optimization core is `src/models/algorithm_2b_mahalanobis.py`; the locked experiment drivers are `scripts/run_shuffled_marginals_taskA_experiment.py` and `scripts/run_shuffled_marginals_es_pt_fr`